# VITALS

AI-Powered Chronic Disease Risk Prediction System

## 1. Project Overview

VITALS is a full-stack web application that uses machine learning to predict chronic disease risk based on physiological and behavioral health indicators. It combines multiple clinical datasets with real behavioral survey data (BRFSS) to provide early risk detection across three tiers: Low, Moderate, and High.

## 2. Datasets Used

### 2.1 Heart Disease UCI (heart.csv)

- Source: UCI Machine Learning Repository (Cleveland database)
- Records: ~303
- Provides: age, gender, blood pressure (trestbps), cholesterol (chol), target (heart disease presence)
- Missing features: BMI, glucose — estimated using clinical correlations
- Missing behavioral: smoking (proxied via fasting blood sugar fbs), physical activity (default=2), alcohol (default=0)

### 2.2 PIMA Diabetes (diabetes.csv)

- Source: National Institute of Diabetes and Digestive and Kidney Diseases
- Records: ~768
- Provides: glucose, blood pressure, BMI, age, target (diabetes outcome)
- Missing features: cholesterol — estimated from glucose (correlation ~0.5)
- Limitation: All female patients (gender=0)
- Data cleaning: Replaced biological zeros (0 BMI, 0 BP, 0 Glucose) with median values

### 2.3 Stroke Dataset (healthcare-dataset-stroke-data.csv)

- Source: Kaggle
- Records: ~5,110
- Provides: age, gender, BMI, glucose (avg_glucose_level), smoking_status, target (stroke)
- Missing features: blood pressure (estimated from age + glucose), cholesterol (estimated from glucose)
- Physical activity: mapped from work_type (children=4, Govt_job=3, Private=2, Self-employed=1, Never_worked=0)

### 2.4 NHANES (nhanes_cleaned.csv)

- Source: National Health and Nutrition Examination Survey (CDC)
- Records: Varies

• Provides: age, BMI, blood pressure, glucose, target

• Missing features: cholesterol (estimated), gender (default=1), all behavioral (default=0/2)

### 2.5 Cardiovascular (cardio.csv → cardio_cleaned.csv)

• Source: Kaggle (cardiovascular disease dataset)

• Records: ~70,000

• Provides: age (converted from days), BMI (computed from height/weight), blood pressure (ap_hi), cholesterol (mapped: 1→190, 2→240, 3→300 mg/dL), target

• Missing features: glucose (estimated from age + cholesterol), all behavioral (default=0/2)

### 2.6 BRFSS 2022 (LLCP2022.XPT → brfss_cleaned.csv)

• Source: CDC Behavioral Risk Factor Surveillance System (2022 survey)

• Original records: ~445,000 → Sampled to 50,000 to prevent data dominance

• Provides: age (_AGEG5YR), gender (_SEX), smoking_status (_SMOKER3), physical_activity (_TOTINDA/EXERANY2), alcohol_intake (ALCDAY4), target (composite from diabetes/heart disease/stroke)

• Missing features: BMI (from _BMI5), blood pressure/cholesterol/glucose (estimated from age + BMI)

• Why sampled: 445k BRFSS records would dominate the ~76k total from other datasets, biasing the model toward behavioral-only patterns

## 3. Feature Schema

### 3.1 Raw Input Features (9)

| Feature | Type | Range | Description |
|---|---|---|---|
| age | Numerical | 1–100 | Patient age in years |
| gender | Categorical | 0/1 | 0=Female, 1=Male |
| bmi | Numerical | 15–60 | Body Mass Index (kg/m²) |
| blood_pressure | Numerical | 80–200 | Systolic blood pressure (mmHg) |
| cholesterol | Numerical | 120–350 | Total cholesterol (mg/dL) |
| glucose | Numerical | 60–280 | Fasting glucose (mg/dL) |
| smoking_status | Categorical | 0/1/2 | 0=Never, 1=Former, 2=Current |
| physical_activity | Categorical | 0–4 | 0=Sedentary, 1=Low, 2=Moderate, 3=Active, 4=Very Active |
| alcohol_intake | Categorical | 0/1/2 | 0=None, 1=Moderate, 2=High |

### 3.2 Engineered Features (12 derived, 22 total)

| Feature | Formula | Rationale |
|---|---|---|
| age_group | 0(≤30), 1(≤45), 2(≤60), 3(>60) | Age stratification for risk tiers |

| | | |
|---|---|---|
| bmi_category | 0(≤18.5), 1(≤25), 2(≤30), 3(>30) | WHO BMI classification |
| bp_category | 0(≤90), 1(≤120), 2(≤140), 3(>140) | Hypertension staging |
| glucose_category | 0(≤70), 1(≤100), 2(≤126), 3(>126) | Diabetes risk classification |
| pulse_pressure | BP - (BP × 0.6) | Arterial stiffness indicator |
| metabolic_risk | age×0.25 + bmi×0.20 + bp×0.20 + glucose×0.20 + chol×0.15 | Composite metabolic syndrome score |
| bmi_glucose _interaction | BMI × glucose / 1000 | Insulin resistance proxy |
| age_bp_interaction | age × BP / 1000 | Age-related cardiovascular risk |
| smoking_bp_risk | smoking_status × BP / 140 | Smoking amplifies BP risk |
| smoking_glucose _risk | smoking_status × glucose / 100 | Smoking affects glucose metabolism |
| activity_bmi_risk | (4 - activity) × BMI / 30 | Inactivity amplifies BMI risk (inverse) |
| activity_glucose _risk | (4 - activity) × glucose / 100 | Inactivity amplifies glucose risk (inverse) |
| alcohol_liver_risk | alcohol_intake × glucose / 100 | Alcohol affects liver glucose regulation |

# 4. ML Models

## 4.1 Ensemble Architecture: Soft Voting Classifier

4 models combined with weighted soft voting (weights: XGBoost=3, RF=2, GB=2, LR=1):

### *XGBoost (Weight: 3 — highest)*

**Why:** Best-in-class for tabular data, handles non-linear relationships, built-in regularization.

Hyperparameters: 400 estimators, max_depth=6, learning_rate=0.05, subsample=0.85, colsample=0.85, scale_pos_weight for class imbalance, gamma=0.1, reg_alpha=0.05, reg_lambda=1.0

Role: Primary predictor, captures complex feature interactions

### *Random Forest (Weight: 2)*

**Why:** Robust to outliers, reduces overfitting via bagging, handles mixed feature types.

Hyperparameters: 300 estimators, max_depth=12, min_samples_split=5, class_weight='balanced'

Role: Stable baseline, reduces variance

### *Gradient Boosting (Weight: 2)*

**Why:** Sequential error correction, strong on imbalanced datasets.

Hyperparameters: 250 estimators, learning_rate=0.08, max_depth=5, subsample=0.8

Role: Captures patterns XGBoost may miss

### *Logistic Regression (Weight: 1)*

**Why:** Linear baseline, calibrated probabilities, interpretable.

Hyperparameters: StandardScaler pipeline, C=1.0, class_weight='balanced', solver=lbfgs

Role: Probability calibration anchor, prevents ensemble from being too extreme

## 4.2 Why Ensemble (Soft Voting)?

- Soft voting averages predicted probabilities (not hard labels), giving calibrated risk scores
- Reduces individual model biases — no single model dominates
- Weighted voting gives more influence to stronger models (XGBoost)
- 5-fold stratified cross-validation ensures robustness

## 4.3 Performance Metrics

| Metric | Value |
|---|---|
| Accuracy | 83.4% |
| ROC-AUC | 0.924 |
| F1 Score | ~0.85 |
| CV ROC-AUC | ~0.92 ± 0.01 |
| CV Accuracy | ~0.83 ± 0.01 |

## 4.4 Threshold Optimization

- Default threshold (0.5) is not optimal for health screening
- Scanned thresholds from 0.05 to 0.50 in 0.02 steps
- Selected threshold that maximizes F1 score (favors sensitivity — better for medical screening where false negatives are costly)

# 5. Risk Classification

| Probability Range | Risk Level | Classification | Color |
|---|---|---|---|
| < 0.15 | Low Risk | Healthy | Green |
| 0.15 – 0.35 | Moderate Risk | At Risk | Orange |
| ≥ 0.35 | High Risk | High Risk | Red |

# 6. Tech Stack

## Backend

- Flask (Python) — lightweight REST API
- scikit-learn — model training, preprocessing, evaluation
- XGBoost — gradient boosting classifier
- joblib — model serialization

• flask-cors — cross-origin requests

### Frontend

• React (Vite) — SPA framework
• TailwindCSS — utility-first styling
• Framer Motion — animations
• Chart.js — dashboard charts
• Lucide React — icons

### Data Pipeline

• pandas/numpy — data processing
• BRFSS XPT parser — SAS format reading

## 7. Project Structure

```
Integrate/
■■■ Minor Work/
■   ■■■ preprocess.py          # Unified preprocessing (6 datasets)
■   ■■■ brfss_preprocess.py    # BRFSS-specific preprocessing
■   ■■■ cardio_preprocess.py   # Cardiovascular dataset cleaning
■   ■■■ train_model.py         # Ensemble training + evaluation
■   ■■■ data/                  # Raw + cleaned datasets
■   ■■■ model.pkl              # Trained ensemble model
■   ■■■ model_metadata.json    # Model metrics + threshold
■■■ vitals-app/
■   ■■■ backend/
■   ■   ■■■ app.py              # Flask API (predict endpoint)
■   ■■■ src/
■       ■■■ App.jsx            # Root with state management
■       ■■■ pages/
■       ■   ■■■ Dashboard.jsx  # Model metrics + charts
■       ■   ■■■ Predict.jsx    # Risk assessment form
■       ■   ■■■ Reports.jsx    # Health metrics report
■       ■■■ components/
■       ■   ■■■ layout/        # Sidebar + Topbar
■       ■   ■■■ Hero.jsx       # Landing section
■       ■   ■■■ Features.jsx   # Feature showcase
■       ■   ■■■ HowItWorks.jsx # Process explanation
■       ■   ■■■ ModelInsights.jsx
■       ■   ■■■ Footer.jsx
■       ■■■ hooks/usePrediction.js
■       ■■■ services/api.js
```

## 8. Key Insights

- Multi-dataset fusion overcomes single-dataset limitations — each dataset contributes what it measures best
- Estimated features (cholesterol from glucose, BP from age) use clinically validated correlations, not random values
- BRFSS is the only dataset with real behavioral data — all others default smoking/activity/alcohol to 0/2/0
- Class imbalance handled via scale_pos_weight (XGBoost), class_weight='balanced' (RF, LR), and threshold optimization
- Interaction features (smoking×BP, activity×BMI) ensure behavioral factors influence physiological risk pathways